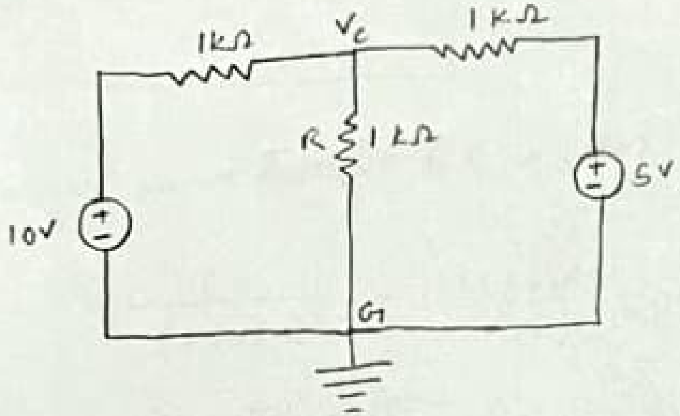
Experiment number - 2

L3-B
Group-12

Pre-Lab work :-

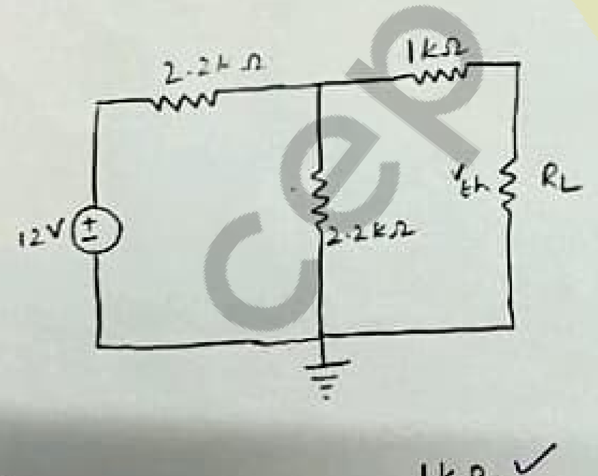
1.



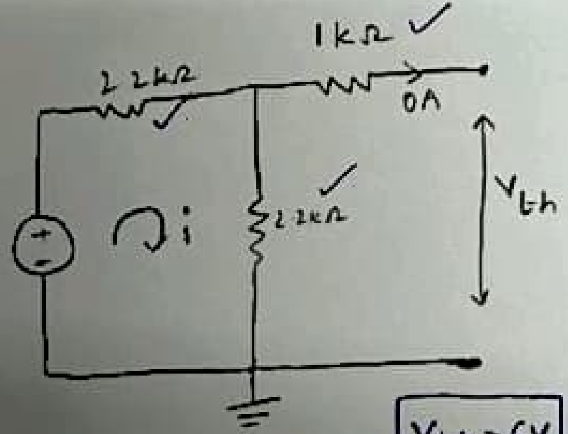
$$\frac{V_c - 10}{1k\Omega} + \frac{V_c - 5}{1k\Omega} = \frac{0 - V_c}{1k\Omega}$$

$$3V_c = 15$$
$$V_c = 5V$$

2.



for V_{th}

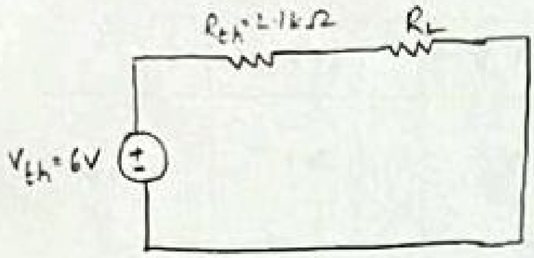


$$V_{th} = \frac{12 \times 2.2k}{(2.2 + 2.2)k} = 6V$$

$$R_{th} = 1k + \left(\frac{2.2 \times 2.2}{2.2 + 2.2}\right)k$$
$$= 2.1k\Omega$$

$\boxed{V_{th} = 6V}$ $\boxed{R_{th} = 2.1k\Omega}$

Equivalent circuit :-



Maximum power transfer to R_L is when $R_L = R_{th}$

$$\Rightarrow i = \frac{6}{(2.1 + 2.1)k\Omega} = 1.428 mA$$

$\therefore$ Power through R_L

$$P_L = i^2 R_L$$

$$= (1.428)^2 \times 2.1$$

$$\boxed{P_L = 4.28 mW}$$

Jayant
4/4/22

PART-A: Superposition Theorem

	V_1 (V)	V_2 (V)	V_A (V)	V_B (V)	V_C (V)
Case-1	10	5	10	5	5
Case-2	10	0	10	0	3.3
Case-3	0	5	0	5	1.6

PART-B: Thevinin's Theorem

V_{OC} (V)	$V_{1\Omega}$ (mV)	V_{Th} (V)	R_{th} (kΩ)
6	28	6	2.14

PART-C: Maximum Power Transfer Theorem

R_L (Ω)	V_A (V)	V_C (mV)	I_L (mA)	Power (mW)
560	1.3	22	2.2	2.81
1000	1.9	19	1.9	3.57
2200	3.2	14	1.4	4.46
3900	3.9	10	1.0	3.89
4700	4.0	9	0.9	3.59

4/4/22

Part-D :-

1) (a) When $V_1 = 10V$, $V_2 = 5V$

we get , $V_{c_1} = 5V$ ...(i)

(b) When $V_1 = 10V$, $V_2 = 0V$

we get , $V_{c_2} = 3.3V$ ...(ii)

(c) When $V_1 = 0V$, $V_2 = 5V$

we get , $V_{c_3} = 1.6V$ ...(iii)

Thus from (ii),(iii) we get

$$V_{c_2} + V_{c_3} = 4.9V$$

So, we have % error $= \frac{5-4.9}{5} \times 100$

$= 2\%$

Hence, we have % accuracy $= 98\%$

Sources of error :-

(i) Error may occur due to loose connection.

(ii) Error may occur due to parallax in oscilloscope reading.

(iii) There may be instrumental error.

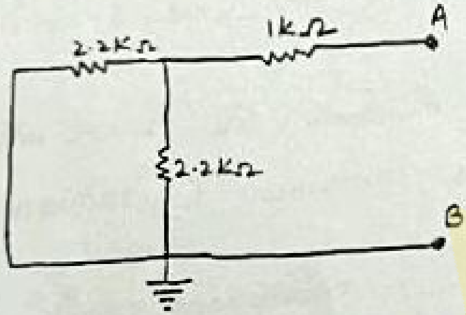
2) $(V_{Th})_{Theoretical} = 6V$

$(V_{Th})_{expt} = 6V$

$(R_{th})_{Theoretical} = 2100\,\Omega$

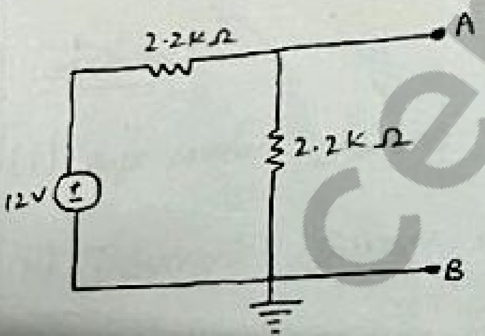
$(R_{th})_{expt} = 2142\,\Omega$

Calculation of R_{th} (Theoretical) :- circuit diagram



$R_{AB} = R_{Th}$ (Theoretical)

Calculation of V_{Th} (Theoretical) :- circuit diagram.



$V_{AB} = V_{Th}$ (Theoretical)

→ No difference in V_{Th} values

→ Difference in R_{Th} values

$$\%\ error = \frac{2100 - 2142}{2100} \times 100 = -2\%$$

Mayank
18/4/22.

Source of error :-

(i) Loose connection

(ii) Parallax error in oscilloscope reading.

(iii) resistance in connecting wires.

(iv) other instrumental error.

3) Maximum power in part(c) is when R_L is nearer to R_{Th} i.e $2200\,\Omega$

This proves the maximum power transfer theorem that power consumed is maximum when $R_L = R_{Th}$

Power maximum :- $R_L = 2200\,\Omega$

$P_{max} = 4.2 \times 10^{-3}\,W$

Conclusion :-

(i) Superposition Theorem is verified ; (Part A)

(ii) Thevenin's Theorem is verified ; (Part B)

(iii) Maximum Power Transfer Theorem is verified ; (Part C)

P_{max} ~~across~~ across R_L is when

$R_L = R_{Th}$

Experiment number-3

L3-B

Group-12

Pre-Lab work :-

1)

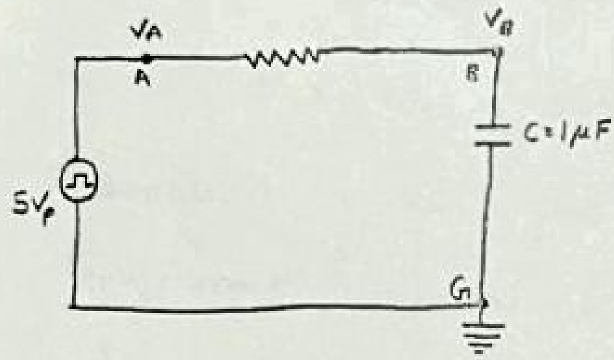


Fig.1

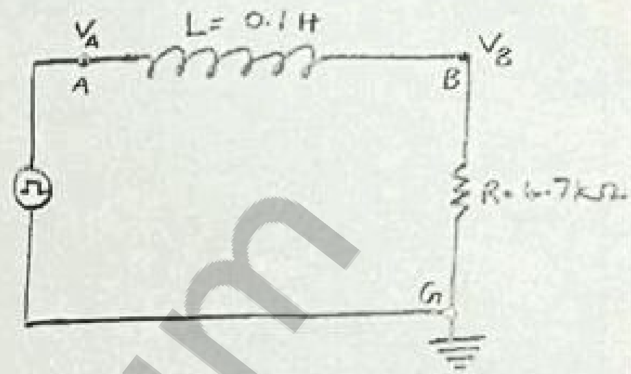


Fig.2

Sol^n:

For Fig.1 → RC Circuit

Time constant $= \tau_1 = RC$

$= (1\times10^{3})(1\times10^{-6})$ sec

$= 10^{-3}$ sec

$= 1$ msec

For Fig.2 ⟶ L·R Circuit

Time constant $= \tau_2 = \frac{L}{R}$

$= \frac{0.1\ H}{4.7\times10^{3}\,\Omega}$

$= \frac{10^{-3}}{47}$ sec

$= 2.13\times10^{-5}$ sec

$= 0.0213$ ms

2)

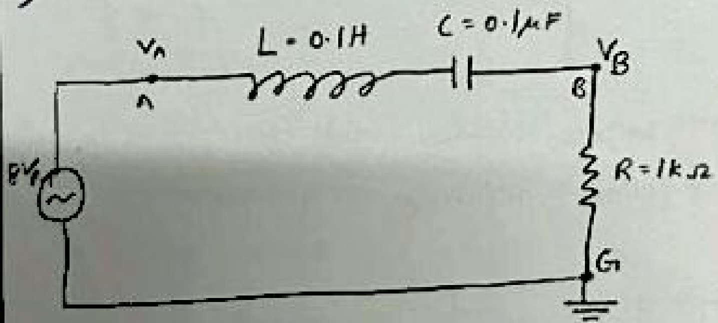


Fig-3

Sol^n: Impedance $= Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$

Resonant frequency $\Rightarrow I_m(z) = 0$

$\omega L - \frac{1}{\omega C} = 0$

$\omega = \frac{1}{\sqrt{LC}}$

$\omega = \frac{1}{\sqrt{(0.1H)(0.1\times10^{-6})F}}$

$= 10^{4}$ rad/s

$\omega = 2\pi f$ $\boxed{f = 1591.55\,Hz}$ $\boxed{\omega = 10^{4}\ rad/sec}$

PART – A : STEP RESPONSE OF RC CIRCUIT

1.

$V_{peak} = 5V$ square wave

frequency $= 0.1 kHz = 100 Hz$

$V_{\tau} = 5 \times 0.63 = 3.15$ volt

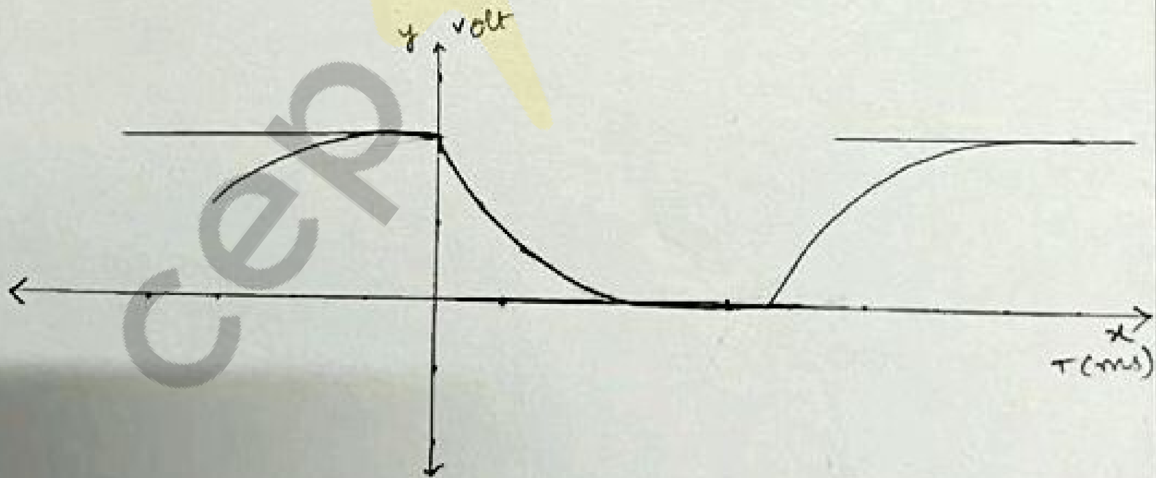
$\tau_{theoretical} = RC = 1000 \times 10^{-6} = 10^{-3} s$ ✓

$\tau_{experimental} = 1ms$ ✓

$T = \frac{1}{f} = \frac{1}{100} = 10 ms$

$\frac{T}{\tau} = \frac{10 ms}{1 ms} = 10$ ✓

2.



3. Frequency for which capacitor just gets charged upto maximum possible voltage $= 100 Hz$ ✓

$\tau = 1ms$

$T = \frac{1}{f} = \frac{1}{1000 Hz} = 10 ms$

$\frac{T}{\tau} = \frac{10 ms}{1 ms} = 10$ ✓

18/4/22.

PART-B: STEP RESPONSE OF RL CIRCUIT

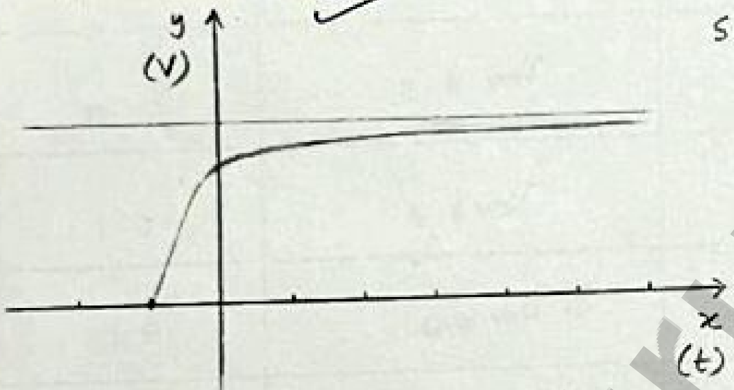
1. $V_{peak} = 5V$ square wave

$frequency = 499.3\ Hz \approx 0.5\ kHz$

$V_{\tau} = 4.98 \times 0.63 \approx 3.14\ V$

$\tau_{theoretical} = \frac{L}{R} = \frac{0.1\ H}{4.7 \times 10^3} = 2.13 \times 10^{-5}\ sec \approx 21\ \mu sec$

$\tau_{experimental} = 20\ \mu sec$



SCALE: X-axis → 1 cm = 20 μs

Y-axis → 1 cm = 2 V

$T = \frac{1}{f} = \frac{1}{499.3} \approx 2\ m\ sec.$

Dayanta
18/4/22.

Part C :- Resonance in series RLC circuit :-

1&2 :-

Frequency (KHz)	voltage across terminal (B-G) (volt)	current in RLC $I = V_{BG}/R{=}1K$
0.1	0.54 volt	0.54mA
0.5	2.4 volt	2.4 mA
1.0	4.6 volt	4.6mA
1.5	6.5 volt ✓	5.5mA
2.0	5.2 volt	5.2 mA
4.0	3.1 volt	3.1mA
10.0	1.3 volt	1.3mA

3. Coarse estimate of the resonant frequency = 1.5 KHz

V_{BG} at $f_{resonance\,(coarse)}$ = 5.5 volt ✓

5. Phase difference at f = 0.1 KHz

nature => ~~laging~~ leading

$$\text{Phase difference} = \frac{2.2\times10^{-3}\times 360}{10^{-4}\times100} = 79.2^{\circ}\ (\text{lag})$$

6. ~~f = 10~~

Phase difference at f = 10 kHz.

nature => leading

$$\text{Phase difference} = \frac{22 \times 10^{-6} \times 360}{0.1 \times 10^{-3}} = 79.2^{\circ} \text{ (lead)}$$

lagging

Dayanand
18/4/22.

Part D :-

1) $\omega_{theoritical} = \frac{1}{\sqrt{LC}} = 10k \text{ rad/s}$

$f_{theoritical} = \frac{1}{2\pi\sqrt{LC}} = 1591.55 \text{ Hz}$

$\omega_{experimental} = 2\pi f_{experimental}$

$f_{experimental} = 1550 \text{ Hz}$

$\omega_{experimental} = 9.73 \text{ krad/s}$.

Error in estimated resonant frequency can occur due to

(i) resistance in connecting wires

(ii) Resistance in instruments used.

(iii) change in resistances due to heating

(iv) loose connections

2) (a) for $\omega = 0.1 \text{ kHz}$

$\omega < \omega_R$

so, $\omega L < \frac{1}{\omega C} \rightarrow X_L < X_C$

$X_C - X_L > 0 \rightarrow$ circuit is capacitive

so current leads voltage in circuit or voltage lags current in circuit

(b) for $\omega = 10 \text{ kHz}$

$\omega > \omega_R$

so $\omega L > \frac{1}{\omega C} \rightarrow X_L > X_C$

$X_L - X_C > 0 \rightarrow$ circuit ~~is~~ is inductive

so current lags voltage or voltage leads ~~circuit~~ current in circuit.

3) At $\omega = \frac{1}{\sqrt{LC}} = 10^4$ rad/s or $f = 1.59$ kHz

Phase difference $\phi = 0$, Net (Impedance) $= Z = R$ $(X_L = X_C)$

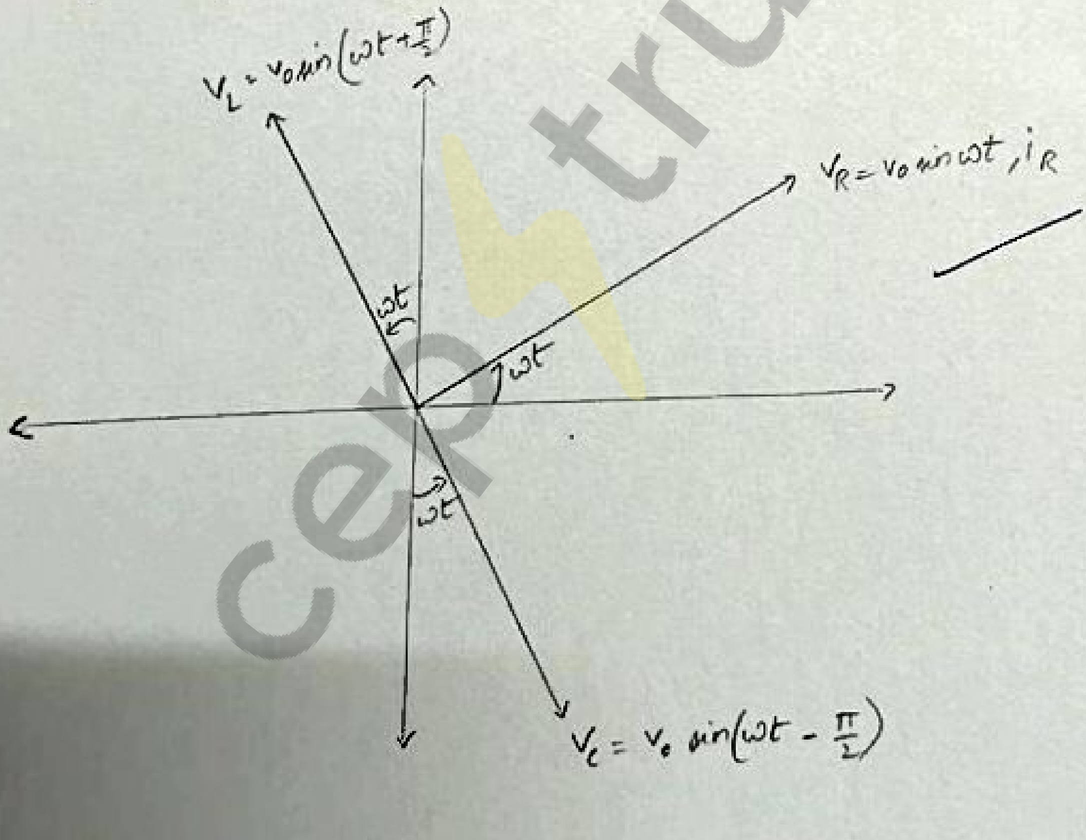
$$V_R = V = V_0 \sin\omega t$$

$$i = \frac{V_0}{R}\sin\omega t = i_R$$

$$V_C = \frac{q}{C} = \frac{V_0}{RC}\int \sin\omega t$$

$$= V_0 \sin(\omega t - \pi/2)$$

$$V_L = L\frac{di}{dt} = \frac{V_0\omega L}{R}\sin\left(\omega t + \frac{\pi}{2}\right) = V_0 \sin(\omega t + \pi/2)$$

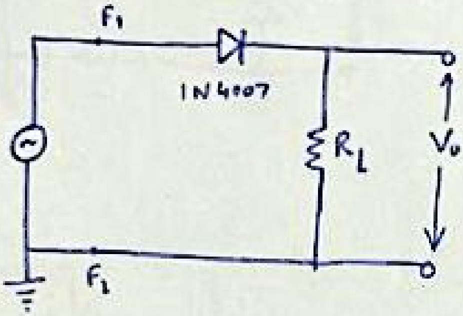


Experiment-4

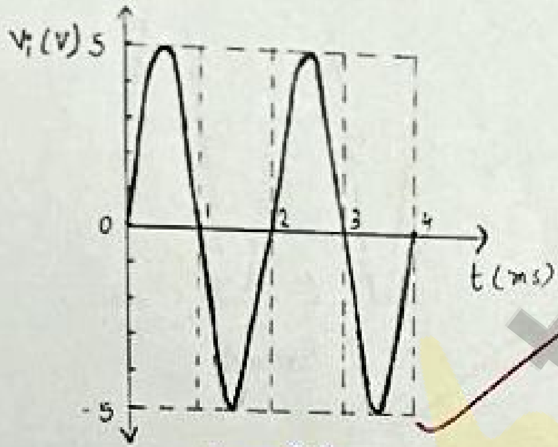
AL3 - B
Group - 12

Pre-Lab

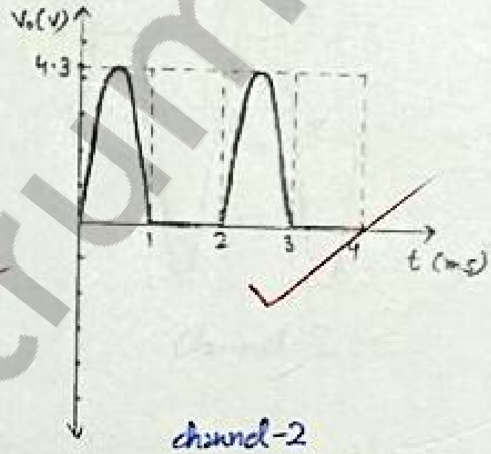
Ans 1.



$V_i = 10\ V\ (pp)$
$f = 500\ Hz$
$R_L = 2.2\,k\Omega$
$V_D = 0.7\,V$
V_o (expected) $= 8.6\ V\ (pp)$

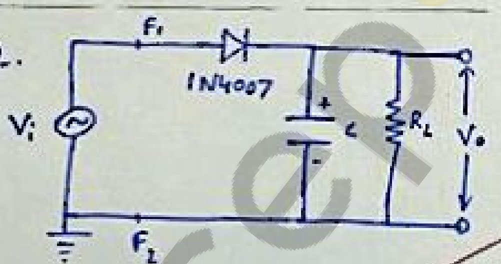


channel 1

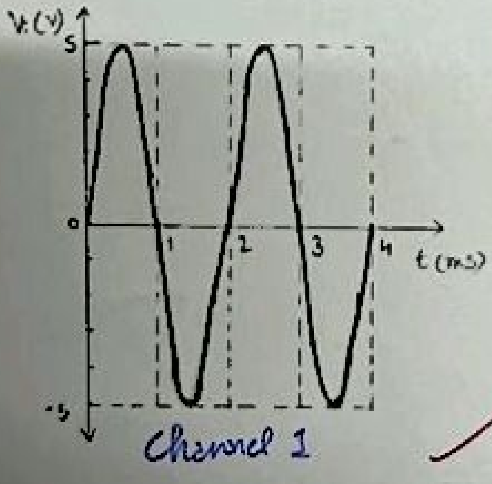


channel-2

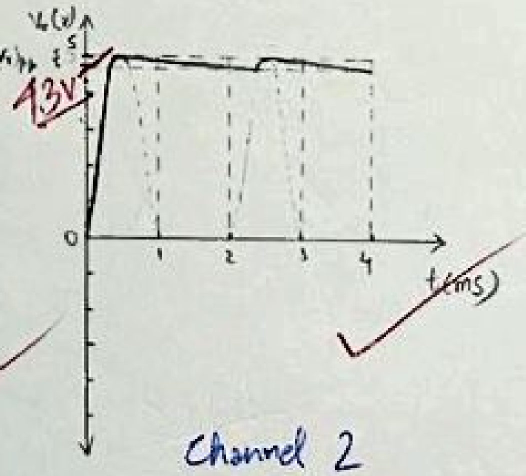
Ans 2.



$V_i = 10\,V\ (pp)$
$f = 500\ Hz \Rightarrow T = 2\,ms$
$R_L = 2.2\ k\Omega\ ,\ C = 22\,\mu F$
$V_{dc} = \left(1 - \frac{T}{2R_L C}\right)5 = 4.896\,V$
$(V_r)_{pp} = \frac{T}{R_L C} \times 5 = 0.208\,V$

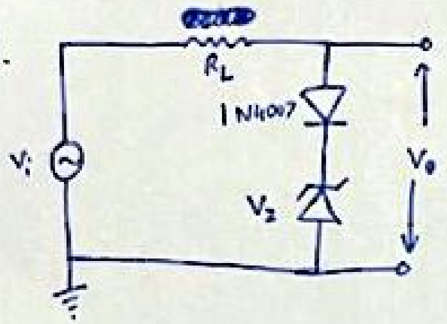


Channel 1



Channel 2

Ans 3.



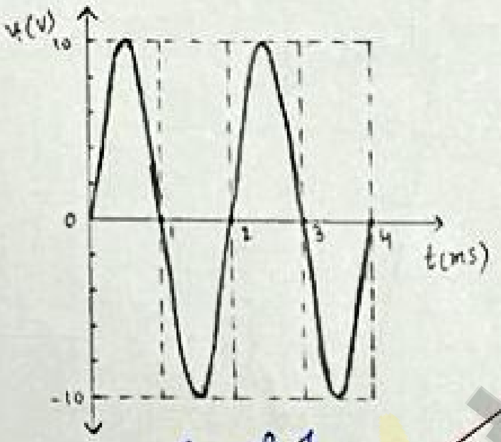
$V_i = 20V$ (pp)

$f = 500Hz$

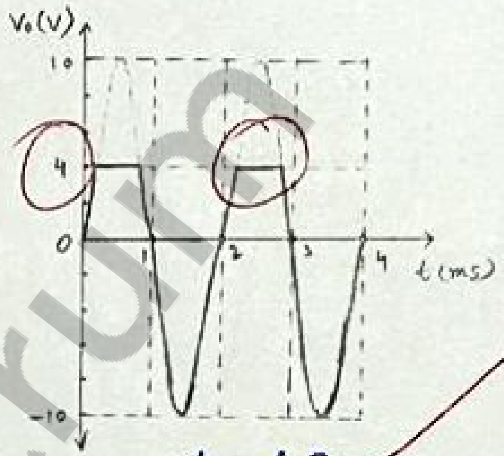
$V_z = 3.9\ V$

$R = 2.2\ k\Omega$

$V_d = 0.7V$



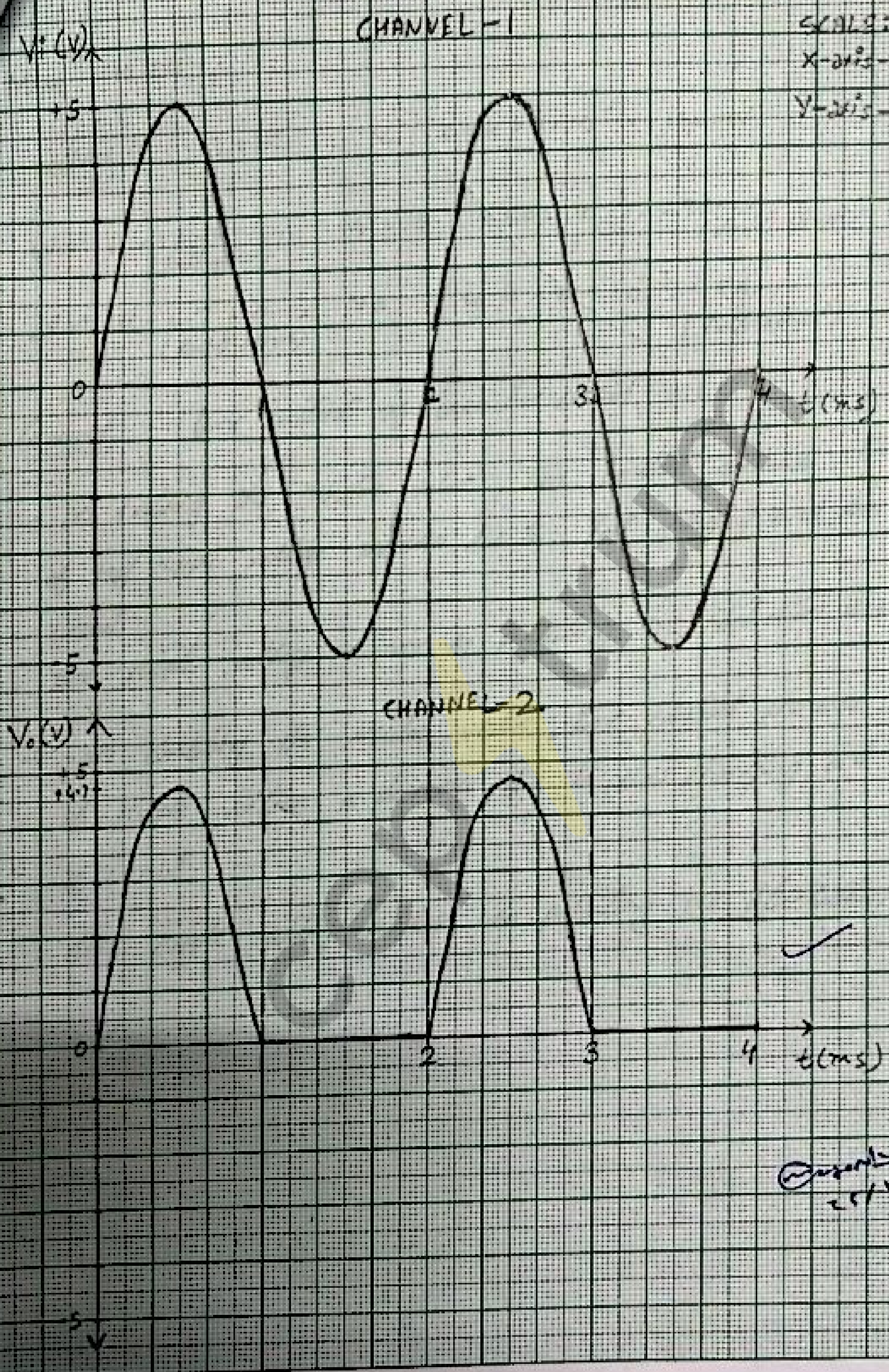
channel 1

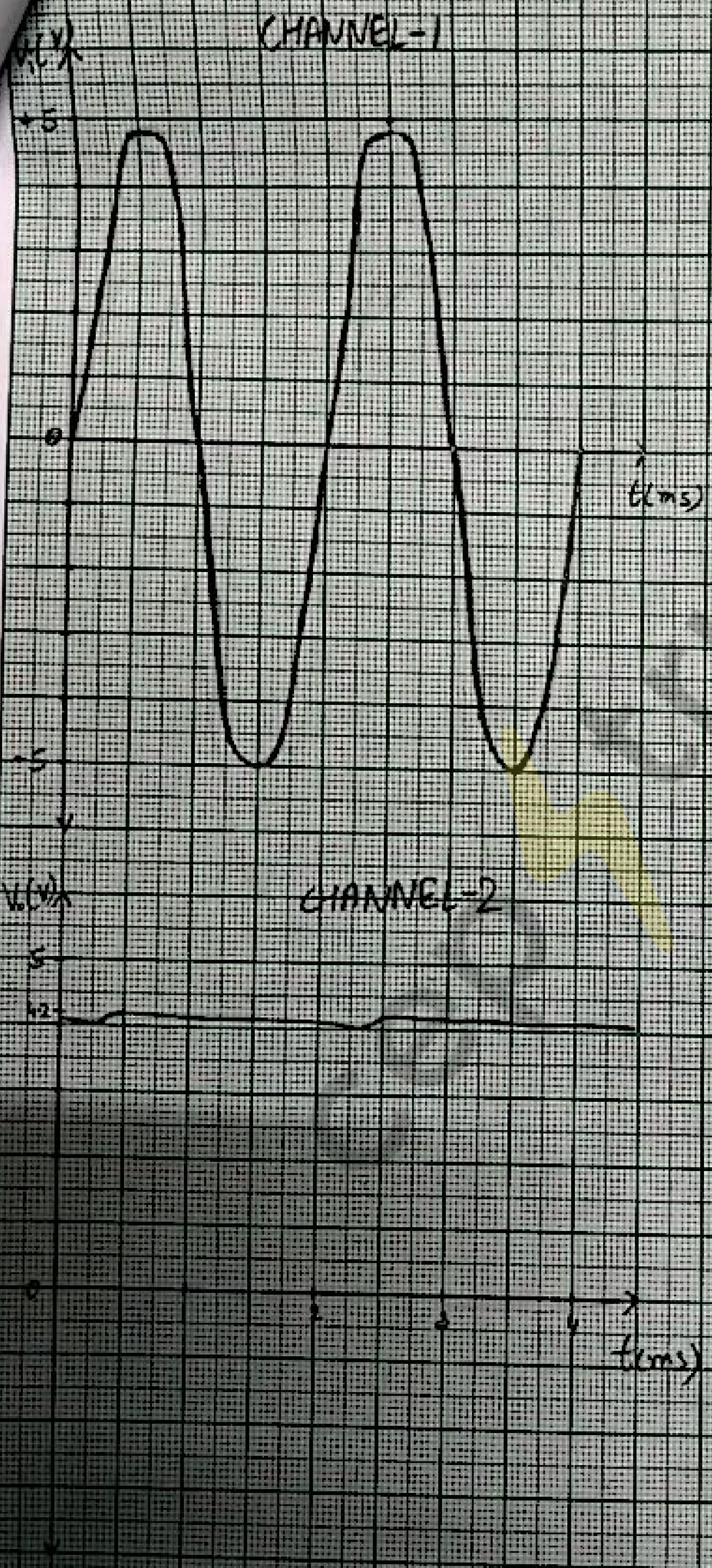


Channel 2

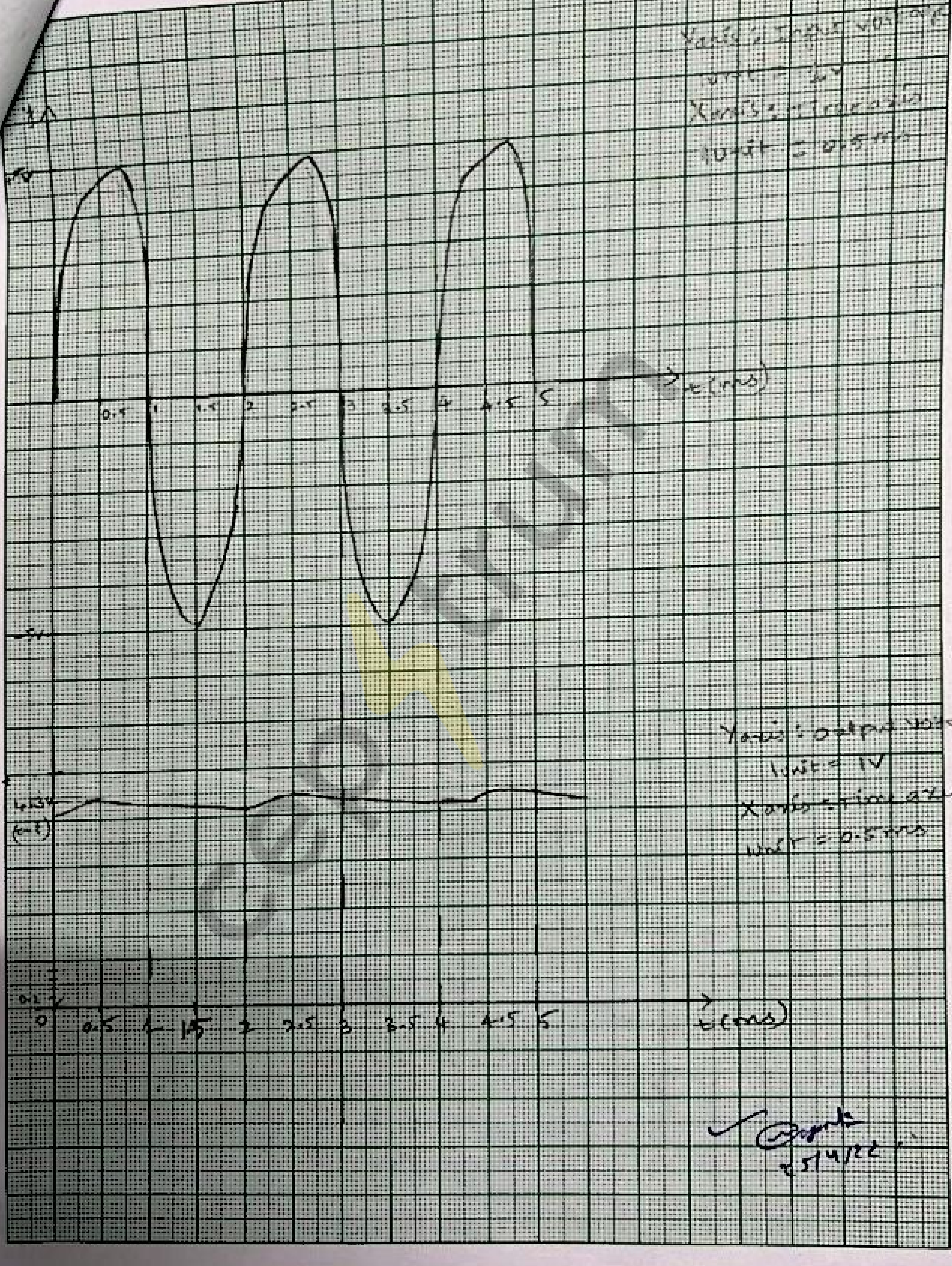
Rahul
25/04/22

SCALE: (For both channels)
X-axis → 1 unit = $\frac{1}{3}$ sec.
Y-axis → 1 unit = 1 Volt

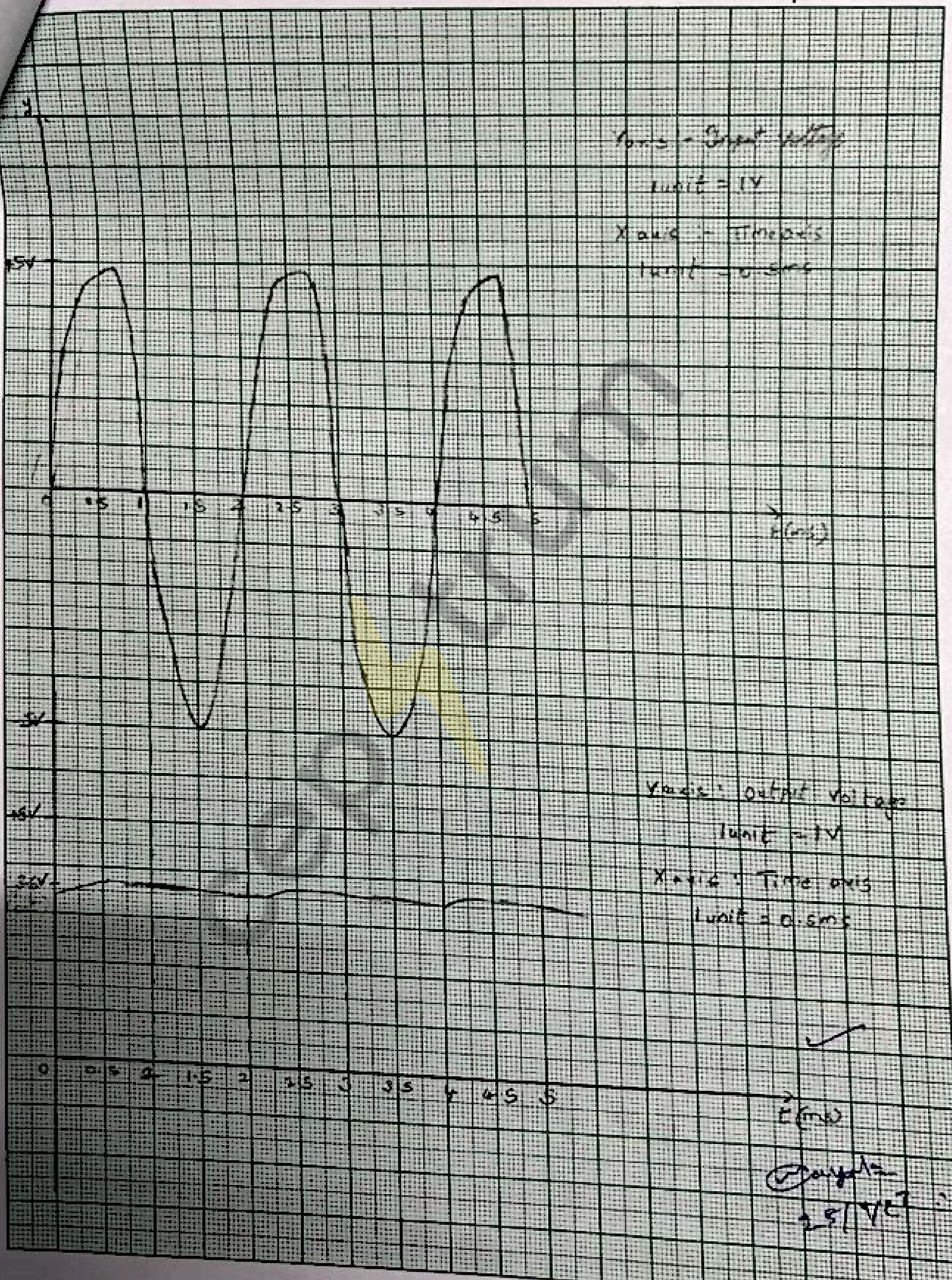




SCALE: (For both channels)
X-axis → 1 unit = 0.5 ms
Y-axis → 1 unit = 1 Volt


Y axis : Input voltage
X axis : time axis
1 unit = 0.5 ms
t (ms)
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
-5V
Y axis : output voltage
1 unit = 1V
X axis : time axis
unit = 0.5 ms
0
t (ms)
5/4/22

YOUVA


Y axis - Input voltage
1unit = 1V
X axis - Time axis
1unit = 0.5ms
Y axis: output voltage
1unit = 1V
X axis: Time axis
1unit = 0.5ms
t(ms)

YOUVA

Part-B :-

V_{input} = 10 volt

when R = 2.2 kΩ , $V_0 = 3.1 \times 50 \times 10^{-3}$ = 155 mV

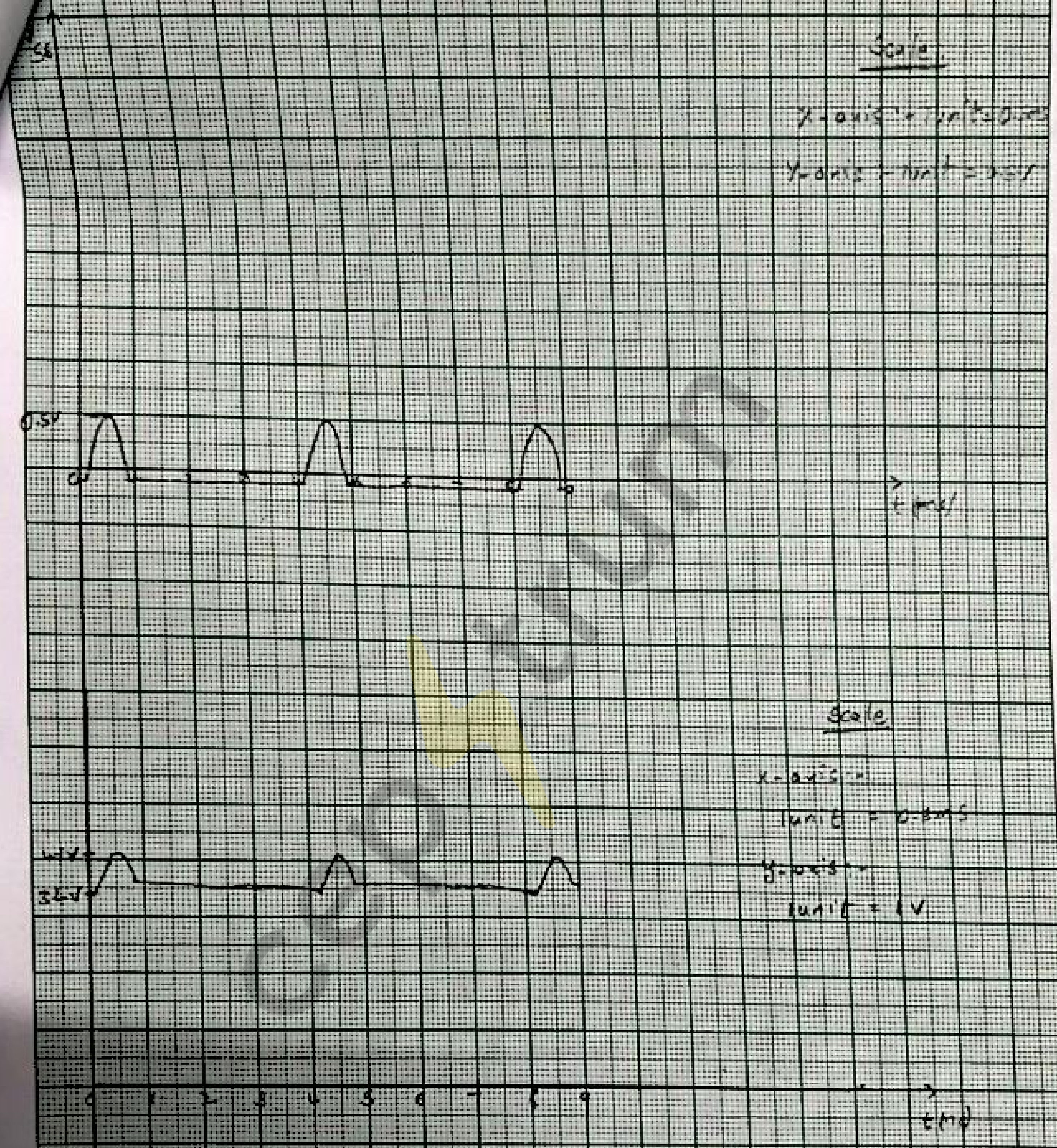
when R = 1 kΩ (half approx) , $V_0 = 5.8 \times 50 \times 10^{-3}$ = 290 mV (double almost)

when R = 4.7 kΩ (double approx) , $V_0 = 4.4 \times 20 \times 10^{-3}$ = 88 mV (half almost)

Comment :- time constant of capacitive circuit = RC

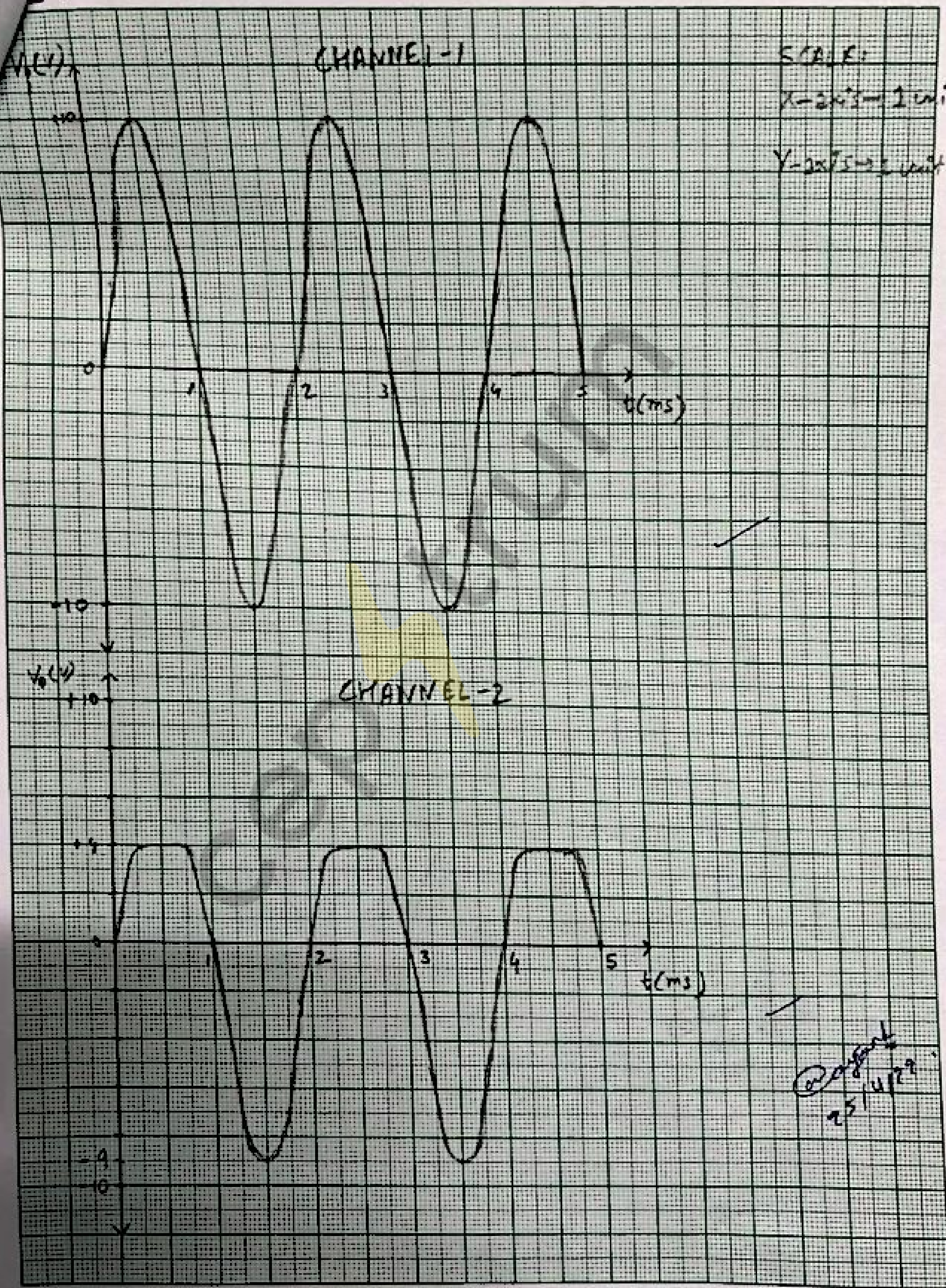
→ on halving the resistor value, time constant decreases, thus amplitude of ~~rupt~~ ripple voltage increases due to fast charging and discharging of circuit.
Here ripple voltage is more so pulsating DC fluctuation is more.

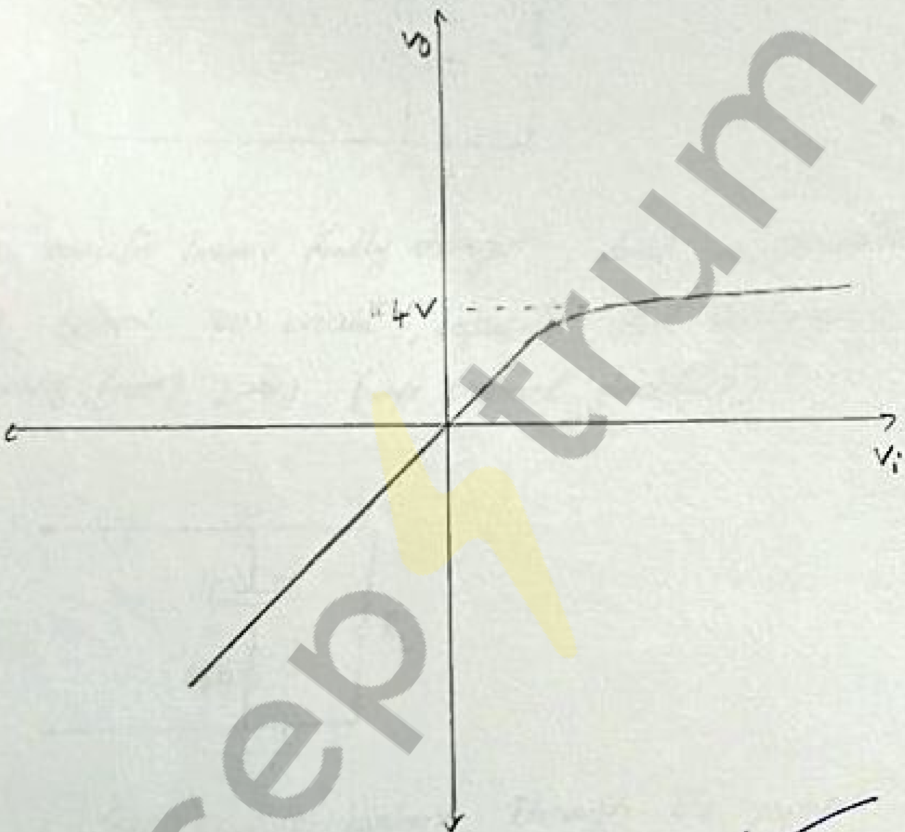
→ on doubling the resistor value, time constant increases, thus amplitude of ripple voltage ~~is~~ decreases due to slow charging and discharging of circuit
Here ripple voltage is less so pulsating DC fluctuation is less.



0.5V
t(ms)
Scale
X-axis 1 unit = 0.5ms
Y-axis 1 unit = 0.5V
Scale
x-axis
1unit = 0.5ms
y-axis
1unit = 1V
0 1 2 3 4 5 6 7 8 9
t(ms)

E :-

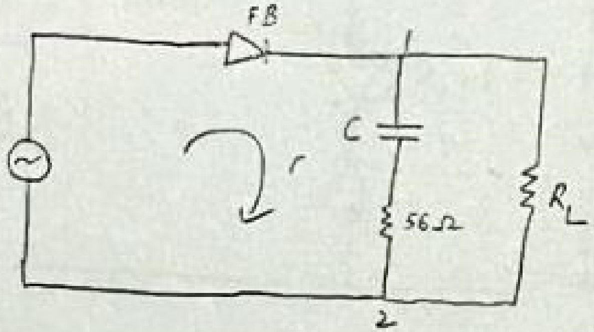




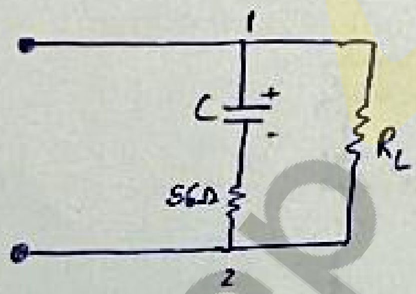
25/4/22.

Part-D:-

① when capacitor is charging, the diode is in forward bias, current is flowing from 1→2 (+ve current direction)

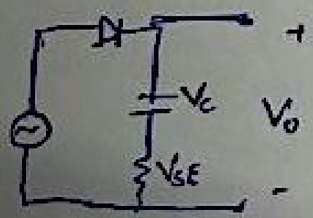


② when capacitor become fully charged, diode is in reverse bias, voltage source become open circuit, capacitor start discharging, so current is flowing from 2→1 (−ve current direction)



So I_C $\left(= V_{56}/56\right)$ → the current through the capacitor, negative for some portion of cycle as shown in part ②.

Estimation of (I_c) peak:-



$V_0 = V_c + V_{56} = i_c \times Z$

$i_c = \frac{V_0}{Z}$, i_c = max, V_0 = max = $(V_i)_{max} - 0.7$

$(V_0)_{max} = 5 - 0.7 = 4.3V$

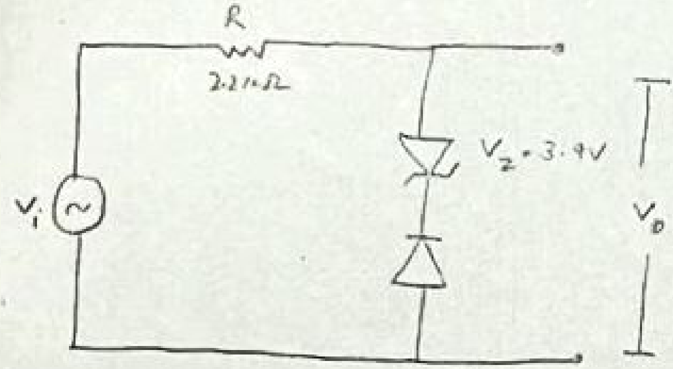
$(V_0)_{max} = 4.3V$ $\quad Z = \sqrt{X_C^2 + R^2}$, $X_C = \frac{1}{j\omega C}$, $\omega = 2\pi \times 500 = 10^3\pi$

$(i_c)_{max} = \frac{4.3}{Z} = 0.0743A$ $\quad Z = 57.84\Omega$ $\quad X_C = 14.46\Omega$ $\quad C = 22\times10^{-6}F$

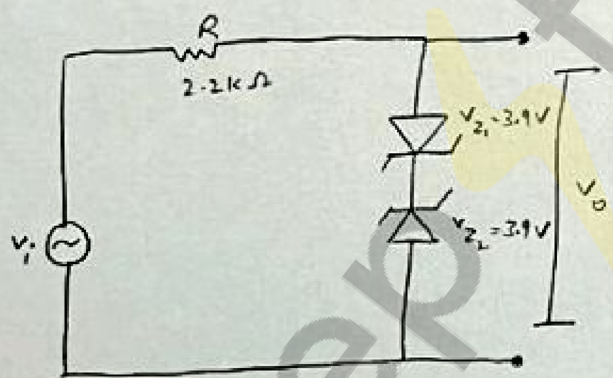
$R = 56\Omega$

$\boxed{(i_c)_{max} = 0.0743A}$

-ve clipper



(b) +ve & -ve clipper

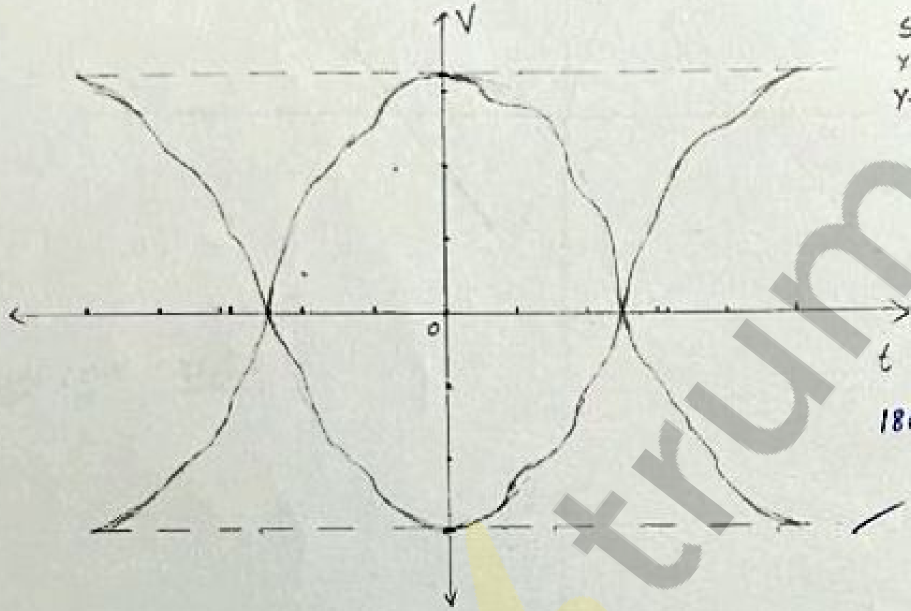


14/5/22

PART - A : Unregulated Power Supply – Full Wave Rectifier (FWR)

1. Transformer Tx → 230V to 12-0-12V, 1A and R_L = 1kΩ rating

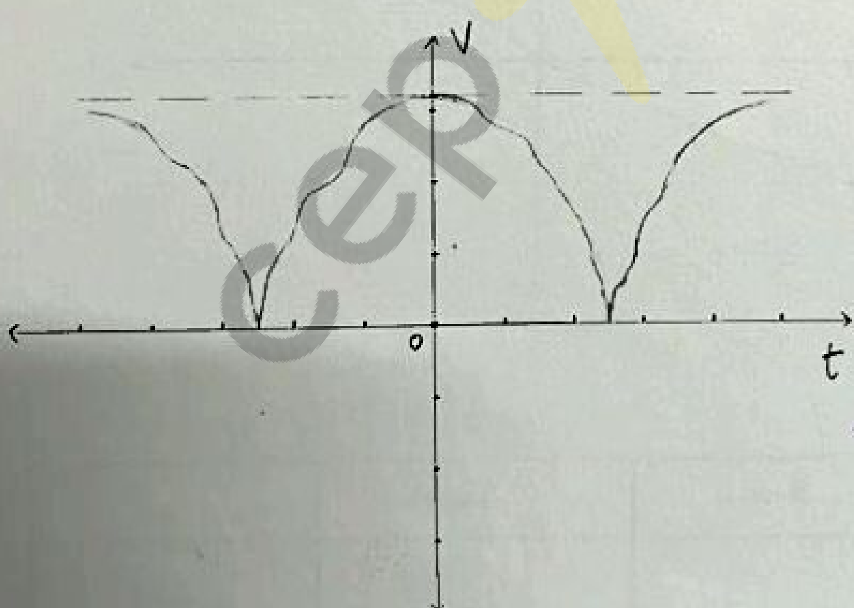
2. V_{AG} = 3×5 = 15V | multiplier = 5
   V_{BG} = 3·2×5 = 16V | time scale = 2ms



SCALE
Y-axis : 1 unit = 2×5 = 10ms
Y-axis : 1 unit = 5V

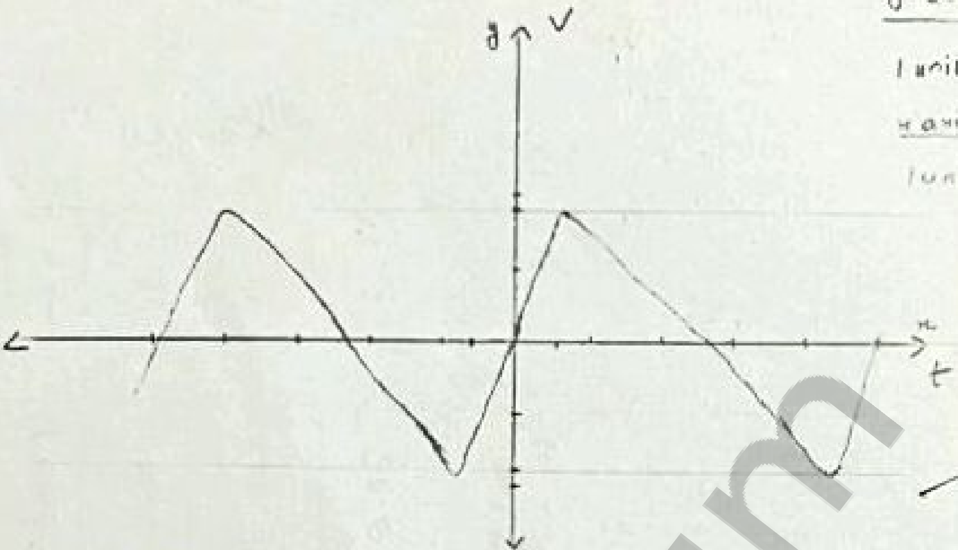
180° out of phase

3.



SCALE
X-axis : 1unit = 2×5 = 10ms
Y-axis : 1unit = 5V

full wave rectifier :-



Scale :-

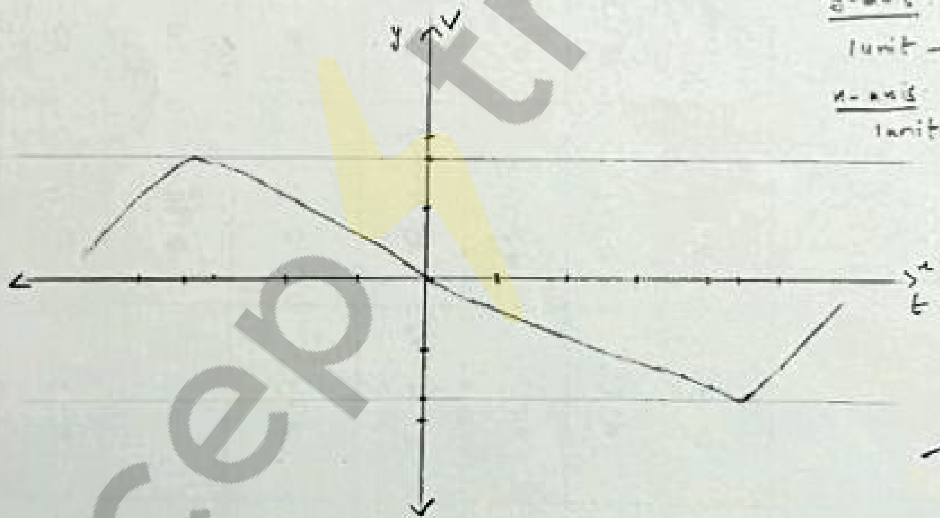
y-axis :-

1 unit → 0.5V/div

x-axis :-

1 unit → 2ms/div

Half wave rectifier :-



Scale :-

y-axis :

1 unit → 1V/div

x-axis

1 unit → 2mS/div

	FWR	H.W.R
V_r	1.7V	3.2V

PART – B : Regulated power supply – zener regulator

R_L	$V_o(max)$	V_r (V)	$V_{oR}(max)$	V_{rR} (V)	V_o (avg)	V_{oR}(avg)
∞	15.5	1.8	6.2	0.11	14.6	6.145
1KΩ	15.5	1.8	6	0.104	14.6	5.948
220Ω	15.5	2	4.2	0.5	14.5	3.95
R_L	I_R (A)	I_L (mA)	I_Z (mA)	P_Z (mW)	P_R (W)	% Reg
∞	0.0150	0	15	92.175	0.127	0
1KΩ	0.0155	5.9	9.6	57.1	0.134	3.2
220Ω	0.0188	17.9	0.9	3.55	0.198	35.72

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Part C :-

1. No, we didn't get the peak voltage in both the halves equal in step 3 of part A.

$V_{AG} = 15V$

$V_{BG} = 16V$

It is because centre tap is not proper

Ripple Frequency in F.W.R $= \frac{1}{3.8 \times 5 \times 10^{-3}}$

$= 52.63\ Hz$

Ripple Frequency in H.W.R $= \frac{1}{5 \times 2 \times 10^{-3}}$

$= 100\ Hz$

2. Yes, I_L have poor regulation it may be due to very small value of R_L. For $R_L = 220\ \Omega$, percentage regulation is very high Thus Thus the poor regulation occur due to very low value of R_L.

3. Given : $I_Z(min) = 5mA$

$P(max) = 0.5$ watt

V_{max} (accross zener) $= \frac{P_{max}}{I_{min}} = \frac{0.5}{5 \times 10^{-3}} = 100V$

$I_R = \frac{V_0 - V_Z}{R} = \frac{13.5 - 6}{960} = 13.9mA$

we can modify the circuit by providing the higher value of R_Z for an even higher I_L $R \times R_L$ are to be formula reduced greatly.

$P_z = \frac{1}{2} W$, $V_z I_z = \frac{1}{2} W$

$I_z = 83.3 mA$

$I_R = I_z(max)$

$R = \frac{V_0 - V_z}{I} = \frac{13.5-6}{83.3} = 108.04 \Omega$

$I_R = I_z(min) + I_z(max) = 83.3 - 5 = 78.3 mA$

4. $P = \frac{V^2}{R} = \frac{(7.5)^2}{108.4} = 520.6 mW$

Yes, we are correct voltage resistance as due to maximum power assumed by resistance was less than power voltage

21/5/22.

Pre-Lab-6

Group: 12

Ans1. Considering ideal op-amp,

$V_+ = V_- = 0$

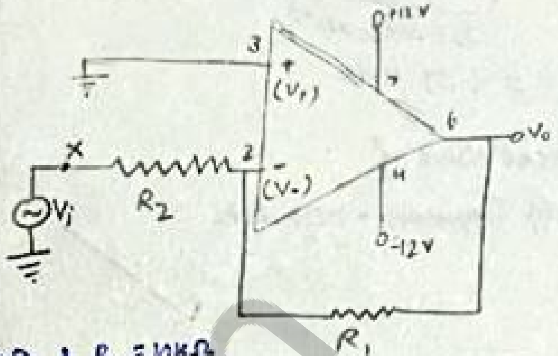
from fig, $V_x = \frac{V_i}{2}$

Applying KVL →

$$\frac{V_0 - V_+}{R_1} = \frac{V_- - V_x}{R_2}$$

$$\Rightarrow \frac{V_0}{R_1} = -\frac{V_i}{2R_2}$$

$$\Rightarrow \boxed{\frac{V_0}{V_i} = -\frac{R_1}{2R_2}}$$

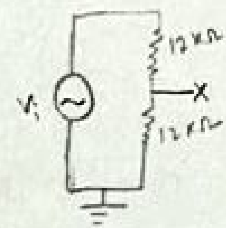


(i) $R_1 = 100\,k\Omega$ & $R_2 = 10\,k\Omega$

$\therefore \frac{V_0}{V_i} = \frac{-100}{2\times10} = -5$

(ii) $R_1 = 10\,k\Omega$ & $R_2 = 1\,k\Omega$

$\therefore \frac{V_0}{V_i} = \frac{-10}{2\times1} = -5$



Ans2. Considering ideal op-amp

$V_+ = V_- = V_x = \frac{V_i}{2}$ (From fig.)

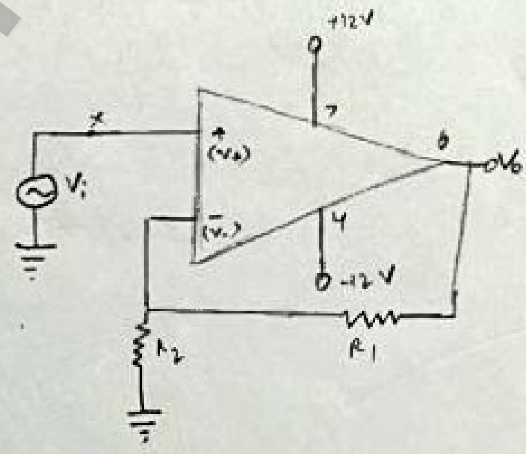
Applying KVL →

$$\frac{V_0 - V_-}{R_1} = \frac{V_- - 0}{R_2}$$

$$\Rightarrow \frac{V_0 - \frac{V_i}{2}}{R_1} = \frac{V_i/2}{R_2}$$

$$\Rightarrow R_2 V_0 = \frac{(R_1 + R_2)}{2} V_i$$

$$\Rightarrow \boxed{\frac{V_0}{V_i} = \frac{R_2 + R_1}{2R_2}}$$



(i) $R_1 = 100\,k\Omega$, $R_2 = 10\,k\Omega$

$\therefore \frac{V_0}{V_i} = \frac{110}{2\times10} = 5.5$

(ii) $R_1 = 10\,k\Omega$, $R_2 = 1\,k\Omega$

$\therefore \frac{V_0}{V_i} = \frac{11}{2\times1} = 5.5$

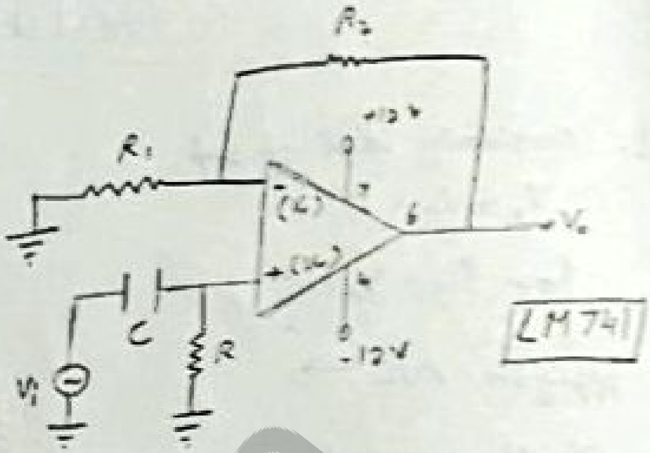
Ans 3. cut-off frequency

$$f_c = \frac{1}{2\pi RC}$$

$$= \frac{1}{2\pi \times 1000 \times 10^{-7}}$$

$$= 1.59 \text{ kHz}$$

∴ Theoretical value of cut-off frequency = 1.59 kHz



21.05.22

PART- A :

Inverting Amplifier :

1. $R_1 = 10k\Omega$ and $R_2 = 1k\Omega$

2. from function generator

$V_i \Rightarrow V_{p\text{-}p} = 200mV$ frequency = 1KHz

3. Voltage Gain → $A = \frac{V_o}{V_i}$

V_i (mV)	V_o (mV)	$A = \frac{V_o}{V_i}$
200	-2100	-10.5
100	-1000	-10
280	-2700	-9.6

phase difference between V_o and V_i = 180° (out of phase)

so gain ⇒ -ve

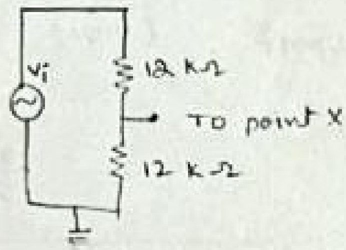
4. change $R_1 = 100k\Omega$ and $R_2 = 10k\Omega$

V_i (mV)	V_o (mV)	$A = \frac{V_o}{V_i}$
104	-1040	-10
200	-2000	-10
280	-2800	-10

5&6:

Keeping V_i at $200\,mV_{P-P}$

first we make potential divider - in circuit

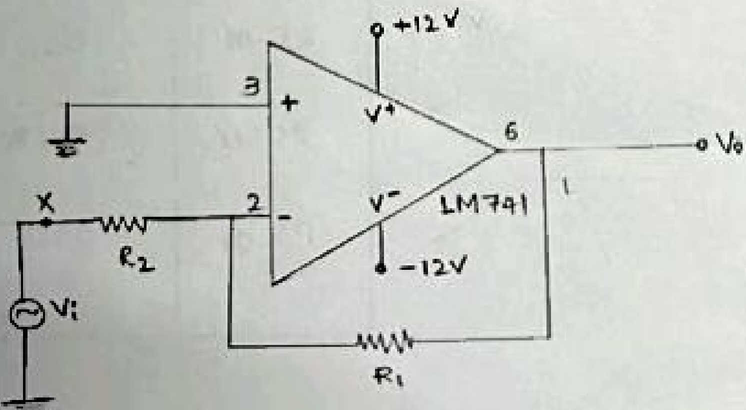


Then $\frac{V_o}{V_i}$ is

(i) $R_1 = 100\,k\Omega$ and $R_2 = 10\,k\Omega$

(ii) $R_1 = 10\,k\Omega$ and $R_2 = 1\,k\Omega$

Resistances	V_i (mV)	V_o (mV)	$A = \frac{V_o}{V_i}$
$R_1 = 100\,k\Omega$ $R_2 = 10\,k\Omega$	200	-620	-3.1
$R_1 = 10\,k\Omega$ $R_2 = 1\,k\Omega$	200	-140	-0.7



For inverting Amplifier

PART - B :

Non-inverting Amplifier

1. $R_1 = 10\,k\Omega$ and $R_2 = 1\,k\Omega$

2. $V_i = 200\,mV$ (V_{p-p}) , frequency = 1 kHz

3. Gain $A = \frac{V_o}{V_i}$

V_i (mV)	V_o (mV)	$A = \frac{V_o}{V_i}$
200	2200	11
108	1160	10.74
290	3200	11.03

phase difference between V_o and V_i = 0°
same phase
so, gain ⇒ +ve

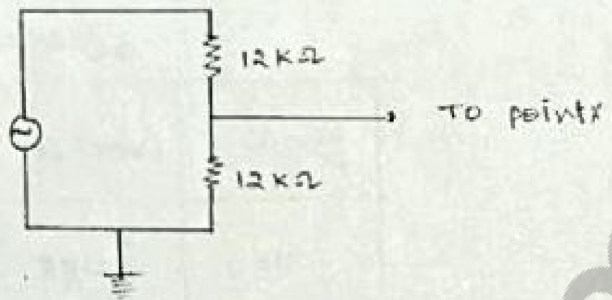
4. change $R_1 = 100\,k\Omega$ and $R_2 = 10\,k\Omega$

V_i (mV)	V_o (mV)	$A = \frac{V_o}{V_i}$
102	1100	10.78
190	2100	11.05
290	3100	10.69

5&6.

$v_i = 200\,mV$ (v_{p-p})

first we make potential divider in circuit

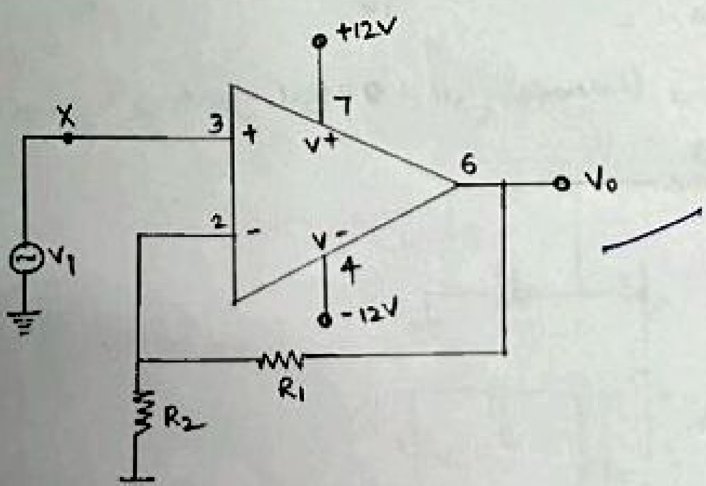


Then we compute $\frac{V_o}{V_i}$

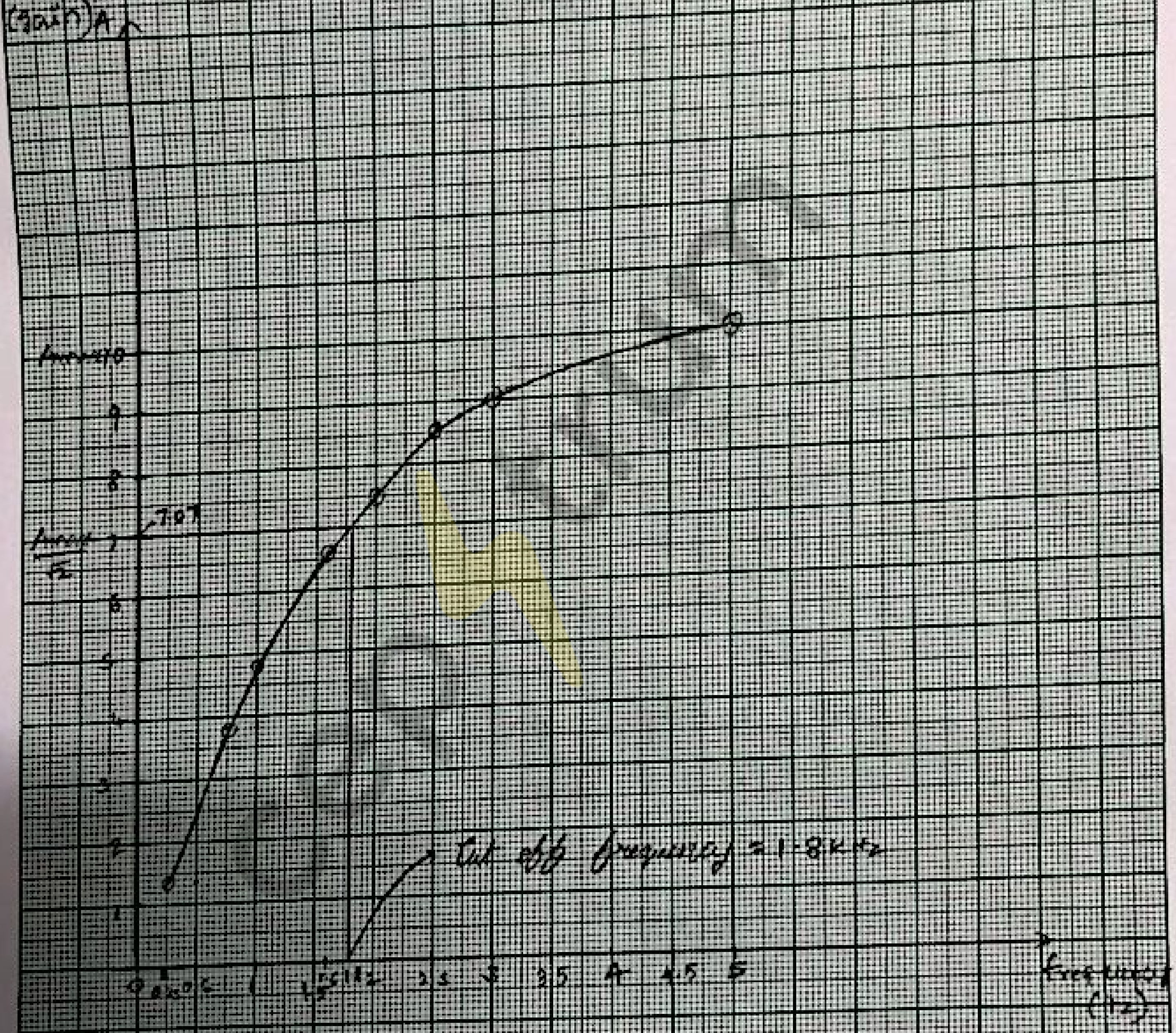
i) $R_1 = 100k\Omega$ & $R_2 = 10k\Omega$

ii) $R_1 = 10k\Omega$ & $R_2 = 1k\Omega$

Resistances	V_i (mV)	V_o (mV)	$A = \frac{V_o}{V_i}$
$R_1 = 100k\Omega$ $R_2 = 10k\Omega$	200	1100	5.5
$R_1 = 10k\Omega$ $R_2 = 1k\Omega$	200	1080	5.4



for Non-inverting Amplifier



For Non-inverting high pass filter

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PART-D :

1. For a source with high internal impedance, a non-inverting op-amp will be suitable.

Reason: for a inverting amplifier :

$$A = -\frac{R_f}{R_{I(I)}}$$

Here, A = gain;

R_f = output impedance

R_i = Input impedance

for a non-inverting amplifier : $A = 1 + \frac{R_f}{R_{I(NI)}}$

Now, in comparing the magnitude of gain, by keeping A & R_f same in both cases,

we see

$$R_{I}(NI) > R_{I}(I)$$

So, we need non-inverting amplifier, for high internal impedance

@ajanta
28/5/20.